

Before: Python for Image and Text Extraction

Using Python for Text Extraction

Working with a Python library (a set of prewritten code that makes it easier to do certain tasks) can be an easy and free way to extract text from images and PDFs. Most of these libraries can be setup quickly and run from either the command line or within a Jupyter notebook. Much like their paid alternatives, these Python libraries make use of the latest AI innovations and Learning Models. If your document images are primarily of modern printed or typed texts, then pyTesseract or EasyOCR might be what you want.

pyTesseract

PyTesseract is an open-source Python library used to extract text from image files. Built as a wrapper for Google's [Tesseract-OCR Engine](#), it relies on traditional image processing techniques and pattern matching for character recognition. It can read all image types supported by the Pillow and Leptonica imaging libraries, including jpeg, png, gif, bmp, tiff, and others. It also has unicode (UTF-8) support and can recognize over 100 languages. PyTesseract performs well on high-resolution, clean images with clear text and simple layouts. Preprocessing (image cleanup, noise reduction, deskewing) is required. Limited support for handwriting transcription.

[pyTesseract Demo Jupyter notebook](#)

[Official pyTesseract Github](#)

EasyOCR

EasyOCR is an open-source Python library (like PyTesseract) used to extract text from image files. EasyOCR leverages deep learning models for text detection and recognition. It excels at recognizing text in challenging conditions: noisy images, varying fonts, complex layouts, and distorted text. Its deep-learning approach means less extensive image preprocessing is needed. It can also leverage GPU acceleration for faster processing. EasyOCR supports 80+ languages and popular writing scripts, including Latin, Chinese, Arabic, Devanagari, Cyrillic, etc. Currently weak on handwriting transcription, but this is next in the pipeline.

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[EasyOCR Demo Jupyter notebook](#)
[Official EasyOCR Github](#)

1. Image Extraction Paid Tools

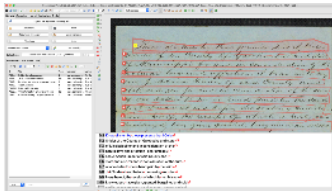
Paid Tools for Text Extraction

Transkribus

<https://www.transkribus.org/>

Transkribus is a powerful AI-based transcription software designed to convert scanned handwritten and print documents into editable text. Transkribus offers itself as a comprehensive digital solution (intuitive dashboard and controls, cloud storage, multiple formats). Transkribus is a popular choice for those working with historical documents, especially medieval texts and handwritten manuscripts.

- uses OCR and HTR technology to convert images into text
- can handle complex structures and document layouts
- 300+ public models to handle historical documents
- ability to train new AI models
- smart search across the entire collection



Pricing Tiers: Free (limited to 50 credits per month and reduced feature set); Scholar (19.99€/month or 99€/year); Team (69€/month or 399€/year). 1 credit = 1 handwritten page or 2 printed pages. An organization level license is also available.

Recommendations: For smaller projects, Transkribus may be a great intuitive solution. For projects involving a lot of pages or many documents, it may be better to go with a less expensive or free solution.

Arkindex

<https://teklia.com/our-solutions/arkindex/>

Arkindex is a platform developed by Teklia for the automatic processing of large collections of scanned documents. It can execute any document processing algorithm: OCR, HTR, feature extraction, captioning, translation, etc. Its architecture has been designed to be generic, enabling it to store any type of result, with generic and configurable types.

- can apply OCR and HTR to images
- customizable workflow design
- seamless integration with custom and open source components
- IIIF-based and accessible through REST API
- can host locally or in the cloud.

Pricing Tiers: 1) Arkindex Open-Source (user-managed) is Free for public and private projects. 2) Hosted by TEKLIA cost is dependent on volume. 3) Enterprise Edition (user-managed, but integrated with High-Performance Computing) offers more granular permissions and security for larger projects. Enterprise costs 1500€/month.

Recommendations: Arkindex might be a good solution for those interested in a cost-effective way of working with a large collection of texts or a corpus that spans many pages.

eScriptorium

<https://escriptorium.rich.ru.nl/>

eScriptorium is a web application offering a workplace to manage the various sets of a transcription campaign. Steps can be manual or automatic processes, and can be applied to printed documents or handwritten ones. The application uses Kraken as a segmentation/transcription engine.

- can apply OCR/HTR to images of printed and handwritten documents
- manual transcription through browser interface for editing segmentations and transcriptions

- can create new models or finetune existing ones to improve automatic recognition
- can import and export models and text transcriptions in a variety of formats
- accessible through a full REST API

Pricing Tiers: eScriptorium is provided as Free and Open Source Software, not as a service.

Recommendations: This is another promising free alternative to Transkribus, but it is currently only available by invitation.

MS Azure Document Intelligence

Azure AI Document Intelligence is an AI service that applies advanced machine learning to extract text, key-value pairs, tables, and structures from documents automatically and accurately. Document Intelligence provides prebuilt models that will work for most scenarios, but can also accommodate custom models tailored for specific document layouts (usually for more complex business or scientific texts).

- ready-to-use models for common document types
- can handle complex structure and layout in documents
- can extract text, key-value pairs, and tables
- REST API provided for custom code solutions
- can extract formulae and barcodes

Pricing Tiers: At the free tier, 500 pages. At S0, \$1.50/1000 pages for the first 1 million pages.

Recommendations: A good choice for relatively contemporary documents, including ones with fairly complex layouts. Document Intelligence does a good job of separating different regions of a page, identifying which text elements belong together (eg. for court transcripts, it will isolate the line numbers running down the left margin and keep them separate from the paragraphs of the reported dialog).

[Document Intelligence Guide](#)

AWS Textract

Amazon Textract is a machine learning (ML) service that extracts text, handwriting, layout elements, and data from scanned documents. It goes beyond simple OCR to identify, understand, and extract specific data (usually for business purposes). Relying on pretrained models, Amazon Textract can extract data in minutes instead of hours or days.

Pricing Tiers: Free tier (1000 pages/month for general text) for the first three months. Otherwise, \$1.50/1000 pages for the first 1 million pages in a month.

Recommendations: A good choice for printed texts and more contemporary handwritten texts. Will likely struggle with historical handwritten documents since the training models are largely made up of contemporary documents of a business or technical nature.

Further Reading

- [Using Python for Text Extraction](#)

2. Image to Text Extraction

Image to Text Extraction

What is Text Extraction?

Text extraction is the process of turning images of text into an editable format (usually returned as a .txt file or JSON). Text extraction tools and code libraries rely on Optical Character Recognition (OCR) and Handwritten Text Recognition (HTR) approaches to interpret images of writing. OCR excels at reading images of printed text, whereas HTR can interpret handwritten text. The range of text extraction tools spans full-featured platforms like Transkribus to Python libraries like pyTesseract and EasyOCR. Many modern tools are aided by AI techniques like machine learning and natural language processing.

We use text extraction tools to quickly and accurately extract text from one data format that cannot be easily read or analyzed (images) into a new format that can be analyzed through computational and digital methods and tools. While manual transcription by a skilled human reader tends to be the most accurate means of extracting text, sometimes the volume of scanned images requiring transcription would take too many human work hours to complete in a reasonable time.

Automating this process by using an OCR/HTR tool can help researchers access the text data sets they need in a more timely and efficient manner. Although OCR has been around since the 1960s, the tools and approaches have continued to evolve. Many popular text data extraction tools incorporate new technologies like AI and machine learning to aid in increasing the accuracy of transcription and handling particularly complex situations and layouts.

What Can OCR/HTR Do?

OCR and HTR allow us to convert images of printed and handwritten text into text documents that can be more easily read by both humans and machines. Once converted, the text files can then be analyzed and searched for patterns, recurrences, anomalies, and other statistical and textual features. Usually, text extraction with OCR/HTR is the first step in the process of making a historical text more accessible to researchers, so that further analysis and preservation steps can be taken.

Case 1: Printed or typeset documents

This is the easiest case for most contemporary OCR/HTR tools.

For modern and contemporary texts (1800-present), fonts tend to be legible and consistent, letter shapes are conventional, and page layouts are fairly uniform. Earlier (pre-1800) typeset documents often present challenges and introduce errors due to unusual fonts, inconsistent spelling, and unfamiliar/antiquated letter shapes. The presence of other marks (stamps, marginalia, etc) and/or damage (tears, blotches, hole punches, or gaps) can produce errors and mistranscriptions (eg. mistaking a hole punch for an "O" or a "0"). Typical examples include literary texts, court transcripts, legal documents, academic articles, financial documents, correspondence, newspapers, and posters.

Case 2: Manuscripts and handwritten texts

Historical manuscripts, letters, documents, and other texts written by hand can pose many unique challenges to digital transcription. Some texts might feature multiple scribes with different writing styles. Others feature barely legible handwriting, inconsistent or idiosyncratic spelling, intermixed illustrations or marginalia, or the faded presence of previous texts (palimpsests and recycled paper). Handwritten documents can also have irregular lines of text that are skewed, curved, or otherwise distorted. In these cases, working with a tool or library that is capable of HTR is vital.

Case 3: Illustrations or images containing text

Text can also be extracted from scanned artwork, photography, posters, diagrams, charts, tables, or maps. While most OCR/HTR tools can extract any text located within an image as a text blob, you will need a more advanced tool or approach if you wish to also preserve the arrangement/location of individual text elements (typically preserved by identifying different blocks or regions, then recording both the text and the associated location information in a JSON file).

Popular Text Extraction Tools

Depending on your needs, budget, and the type of texts you are working with, there are a wide variety of OCR/HTR tools available, ranging from free open-source Python libraries like **pytesseract** and **pyEasyOCR** to paid proprietary solutions like Transkribus or [Microsoft Azure's Document Intelligence](#). Free options require some level of Python programming, but tutorials and guides are readily available. You can find our Jupyter notebooks for using [pyTesseract](#) and [EasyOCR](#) on the DiSA GitHub.

Further Reading

- [Using Paid Text Extraction Tools](#)

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